Academic Year: 2025-26
Semester: VI
Class / Branch / Div: TE- IT C
Subject: MAD & PWA Lab
Name of Instructor: Saylee Lapalikar

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Student ID:24204018
Roll No.19
Date of Submission:10/04/26

Experiment 11

Deployment of E-commerce PWA to GitHub Pages

Aim: To deploy an eCommerce Progressive Web App

(PWA) on GitHub Pages, making it accessible online.

Program :

Open terminal:

flutter create ecommerce_pwa

cd ecommerce_pwa

code .

```
PS C:\Users\Lenovo\OneDrive\Desktop\MAD_25_26\Exp-8\flutter_firebase\new_application> flutter create ecommerce_pwa
Creating project ecommerce_pwa...
Resolving dependencies in `ecommerce_pwa`... (2.5s)
Downloading packages...
Got dependencies in `ecommerce_pwa`.
Wrote 130 files.

All done!
You can find general documentation for Flutter at: https://docs.flutter.dev/
Detailed API documentation is available at: https://api.flutter.dev/
If you prefer video documentation, consider: https://www.youtube.com/c/flutterdev

In order to run your application, type:

  $ cd ecommerce_pwa
  $ flutter run

Your application code is in ecommerce_pwa\lib\main.dart.

PS C:\Users\Lenovo\OneDrive\Desktop\MAD_25_26\Exp-8\flutter_firebase\new_application> cd ecommerce_pwa
PS C:\Users\Lenovo\OneDrive\Desktop\MAD_25_26\Exp-8\flutter_firebase\new_application\ecommerce_pwa> cd .
PS C:\Users\Lenovo\OneDrive\Desktop\MAD_25_26\Exp-8\flutter_firebase\new_application\ecommerce_pwa> code .
```

Step 2: Enable Web Support

flutter config --enable-web

flutter devices

```
PS C:\Users\Lenovo\OneDrive\Desktop\MAD_25_26\Exp-8\flutter_firebase\new_applicat
ion\ecommerce_pwa> flutter config --enable-web
>> flutter devices
Setting "enable-web" value to "true".

You may need to restart any open editors for them to read new settings.
Found 3 connected devices:
  Windows (desktop) • windows • windows-x64    • Microsoft Windows [Version
  10.0.26200.8037]
  Chrome (web)      • chrome  • web-javascript • Google Chrome 146.0.7680.167
  Edge (web)        • edge    • web-javascript • Microsoft Edge 146.0.3856.84

Run "flutter emulators" to list and start any available device emulators.

If you expected another device to be detected, please run "flutter doctor" to
diagnose potential issues. You may also try increasing the time to wait for
connected devices with the "--device-timeout" flag. Visit
https://flutter.dev/setup/ for troubleshooting tips.
PS C:\Users\Lenovo\OneDrive\Desktop\MAD_25_26\Exp-8\flutter_firebase\new_applicat
ion\ecommerce_pwa>
```

Step 4: Replace UI with Simple eCommerce Design

```
import 'package:flutter/material.dart';


void main() {
  runApp(MyApp());
}


class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context)
    { return MaterialApp(
      title: 'Ecommerce PWA',
      debugShowCheckedModeBanner: false,
```

```
    home: ProductPage(),
  );
 }
}

class ProductPage extends StatelessWidget {
 @override
 Widget build(BuildContext context)
  { return Scaffold(
    appBar: AppBar(
      title: Text("Ecommerce Store"),
    ),
    body:
      Column( childr
      en: [ Card(
          margin: EdgeInsets.all(10),
          child: ListTile(
            leading: Icon(Icons.shopping_bag),
            title: Text("Shoes"),
            subtitle: Text("₹2000"),
            trailing:
            ElevatedButton( child:
            Text("Buy"), onPressed: () {
                ScaffoldMessenger.of(context).showSnackBar( SnackBar(conten
                  t: Text("Shoes added to cart")),
                );
              },
            ),
          ),
```

```
),
Card(
  margin: EdgeInsets.all(10),
  child: ListTile(
    leading: Icon(Icons.watch),
    title: Text("Watch"),
    subtitle: Text("₹1500"),
    trailing:
    ElevatedButton( child:
    Text("Buy"), onPressed: () {
        ScaffoldMessenger.of(context).showSnackBar( SnackBar(conten
          t: Text("Watch added to cart")),
        );
      },
    ),
  ),
),
Card(
  margin: EdgeInsets.all(10),
  child: ListTile(
    leading: Icon(Icons.backpack),
    title: Text("Bag"),
    subtitle: Text("₹1200"),
    trailing:
    ElevatedButton( child:
    Text("Buy"), onPressed: () {
        ScaffoldMessenger.of(context).showSnackBar( SnackBar(conten
          t: Text("Bag added to cart")),
        );
```

```
            },
          ),
        ),
      ),
    ],
  ),
);
}
}
```

Step 4: Run the App

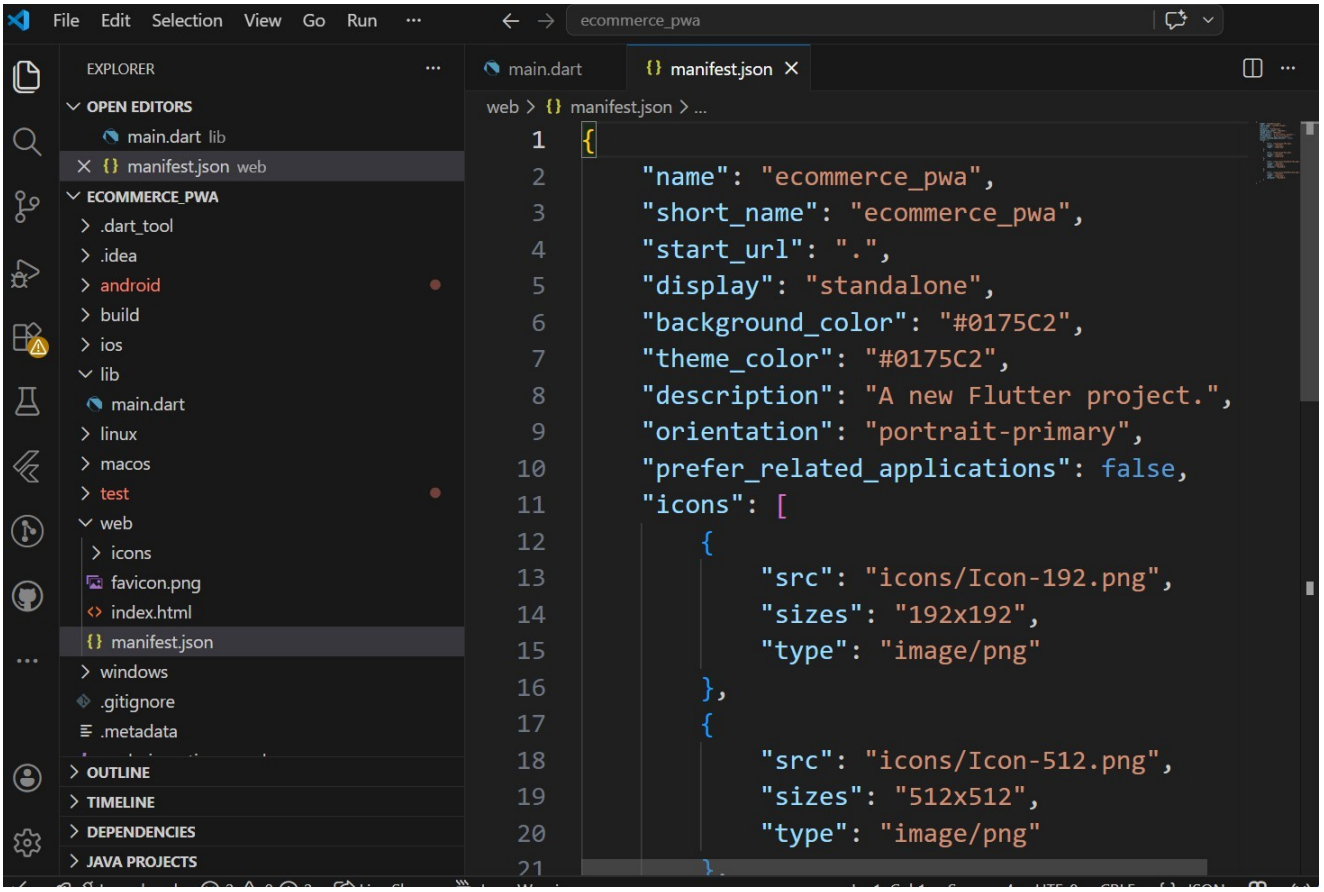
flutter run -d chrome

PART B: Convert to PWA

Step 5: Check PWA Files

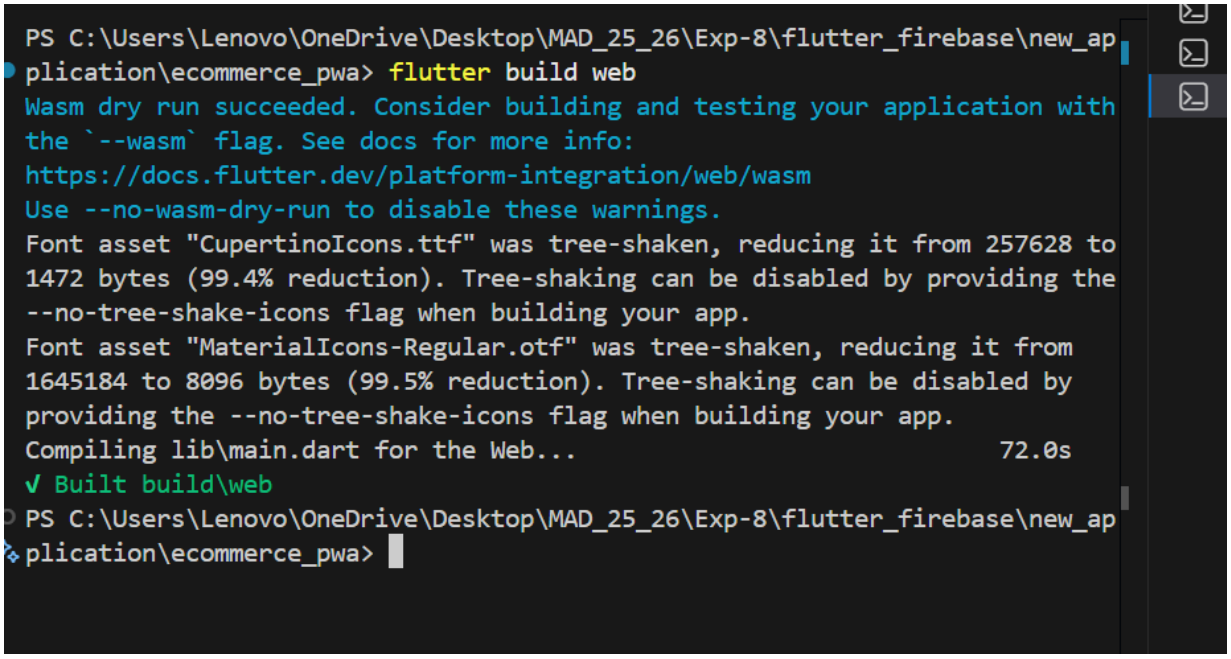
Go to:

- **web/manifest.json**
- **web/index.html**
- **"display": "standalone"**



Step 6: Build Web App

flutter build web



Step 5: Fix for GitHub Pages

flutter build web --base-href "/ecommerce_pwa/"

```
PS C:\ecommerce1_pwa\ecommerce2_pwa> flutter build web --base-href "/ecommerce2_pwa/"
Wasm dry run succeeded. Consider building and testing your application with the `--wasm` flag. See docs for more
info: https://docs.flutter.dev/platform-integration/web/wasm
Use --no-wasm-dry-run to disable these warnings.
Font asset "CupertinoIcons.ttf" was tree-shaken, reducing it from 257628 to 1472 bytes (99.4% reduction).
Tree-shaking can be disabled by providing the --no-tree-shake-icons flag when building your app.
Font asset "MaterialIcons-Regular.otf" was tree-shaken, reducing it from 1645184 to 8096 bytes (99.5% reduction).
Tree-shaking can be disabled by providing the --no-tree-shake-icons flag when building your app.
```

PART C: Deploy to GitHub Pages

Step 8: Initialize Git

git init

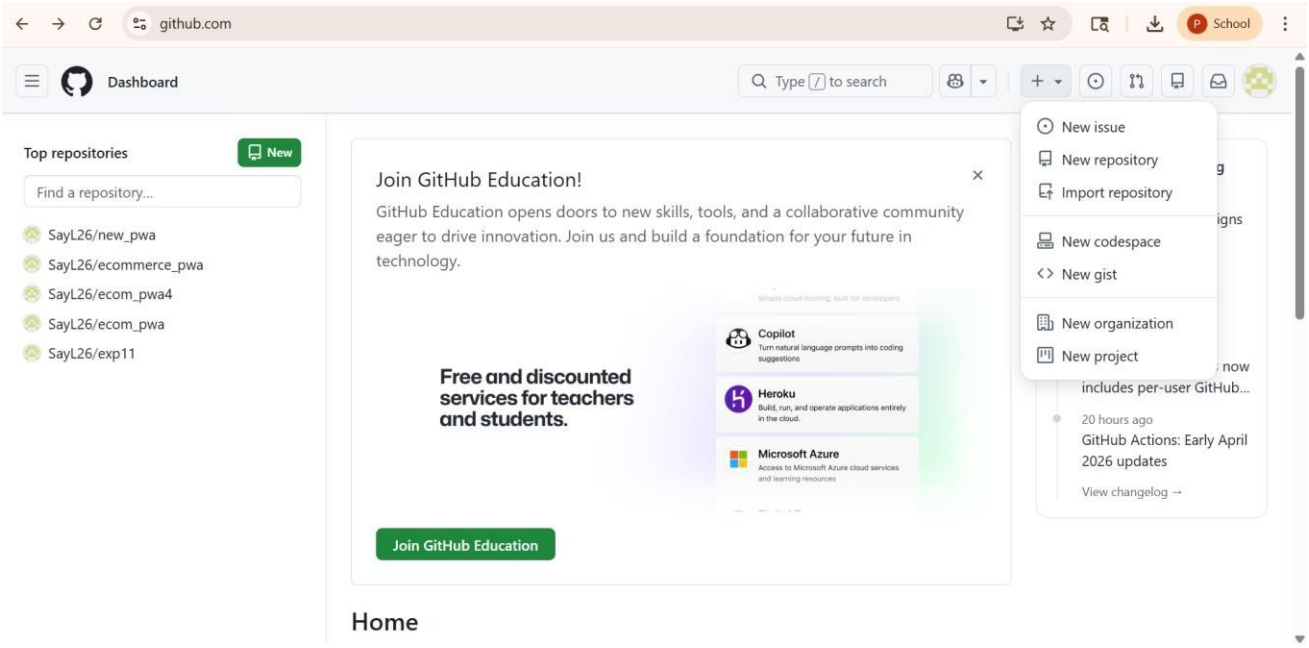
git add .

git commit -m "Initial commit"

```
PS C:\Users\Lenovo\OneDrive\Desktop\MAD_25_26\Exp-8\flutter_firebase\new_application\ecommerce_pwa> git init
>> git add .
>> git commit -m "Initial commit"
warning: in the working copy of 'windows/flutter/generated_plugin_registrant.h', LF will be replaced by CRLF the nex
t time Git touches it
warning: in the working copy of 'windows/flutter/generated_plugins.cmake', LF will be replaced by CRLF the next time
 Git touches it
[master (root-commit) 01f2131] Initial commit
 129 files changed, 4912 insertions(+)
 create mode 100644 .gitignore
 create mode 100644 .metadata
 create mode 100644 README.md
 create mode 100644 analysis_options.yaml
 create mode 100644 android/.gitignore
 create mode 100644 android/app/build.gradle.kts
 create mode 100644 android/app/src/debug/AndroidManifest.xml
 create mode 100644 android/app/src/main/AndroidManifest.xml
 create mode 100644 android/app/src/main/kotlin/com/example/ecommerce_pwa/MainActivity.kt
 create mode 100644 android/app/src/main/res/drawable-v21/launch_background.xml
 create mode 100644 android/app/src/main/res/drawable/launch_background.xml
 create mode 100644 android/app/src/main/res/mipmap-hdpi/ic_launcher.png
 create mode 100644 android/app/src/main/res/mipmap-mdpi/ic_launcher.png
 create mode 100644 android/app/src/main/res/mipmap-xhdpi/ic_launcher.png
 create mode 100644 android/app/src/main/res/mipmap-xxhdpi/ic_launcher.png
 create mode 100644 android/app/src/main/res/mipmap-xxxhdpi/ic_launcher.png
 create mode 100644 android/app/src/main/res/values-night/styles.xml
 create mode 100644 android/app/src/main/res/values/styles.xml
Live Share   Java: Warning   Ln 1, Col 1   Spaces: 4   UTF-8   CRLF   { } JSON   Go Live   No Device   Quokka   Go Live
```

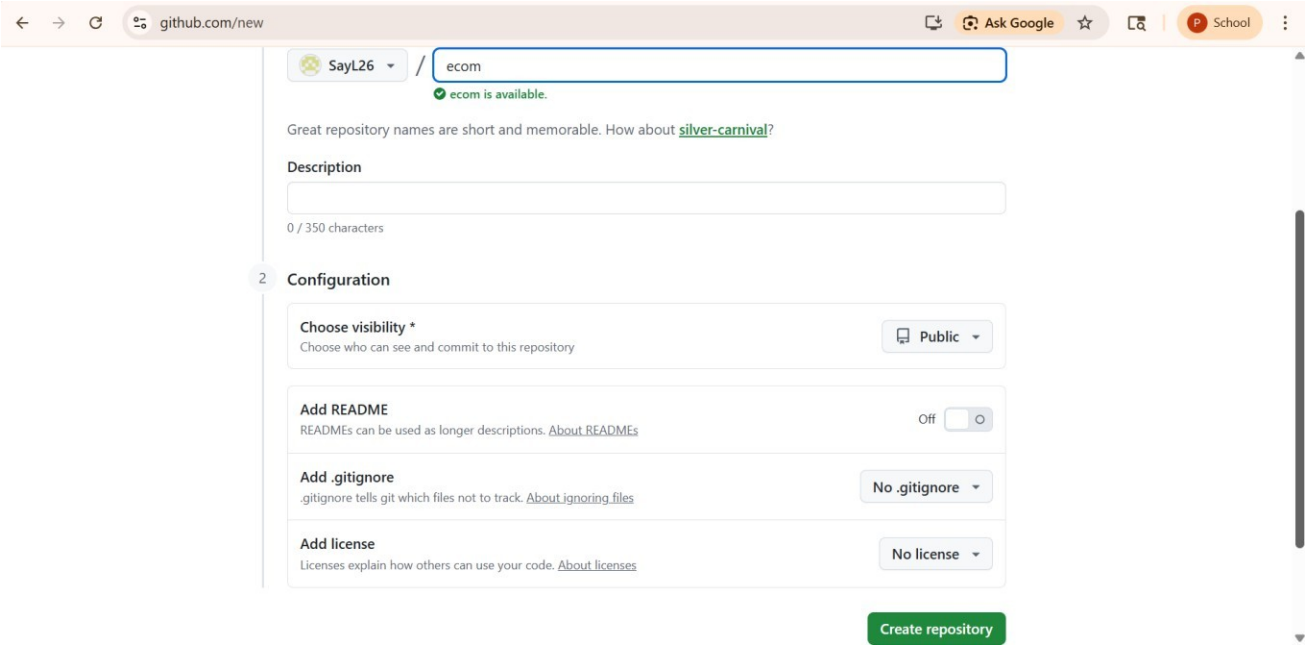
Step 9: Create Repository on GitHub

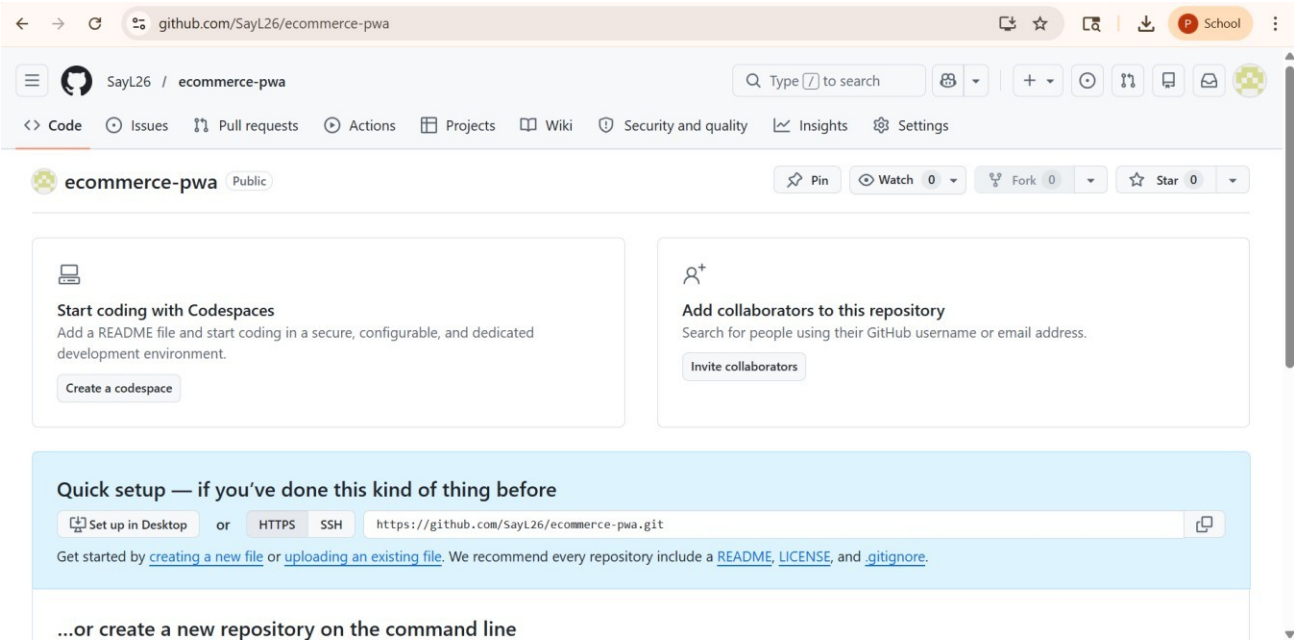
- **Name: ecommerce-pwa**



Step 9: Create Repository on GitHub

- **Name: ecommerce-pwa**





Step 10: Push Code

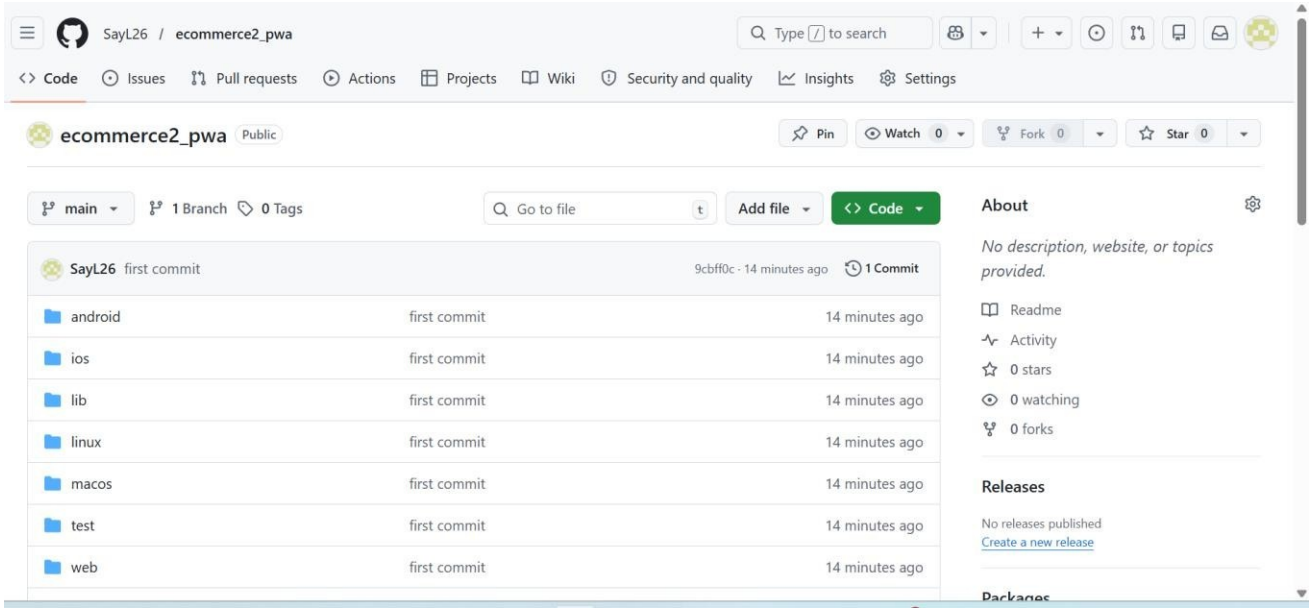
git remote add origin https://github.com/your-username/ecommerce-pwa.git

git branch -M main

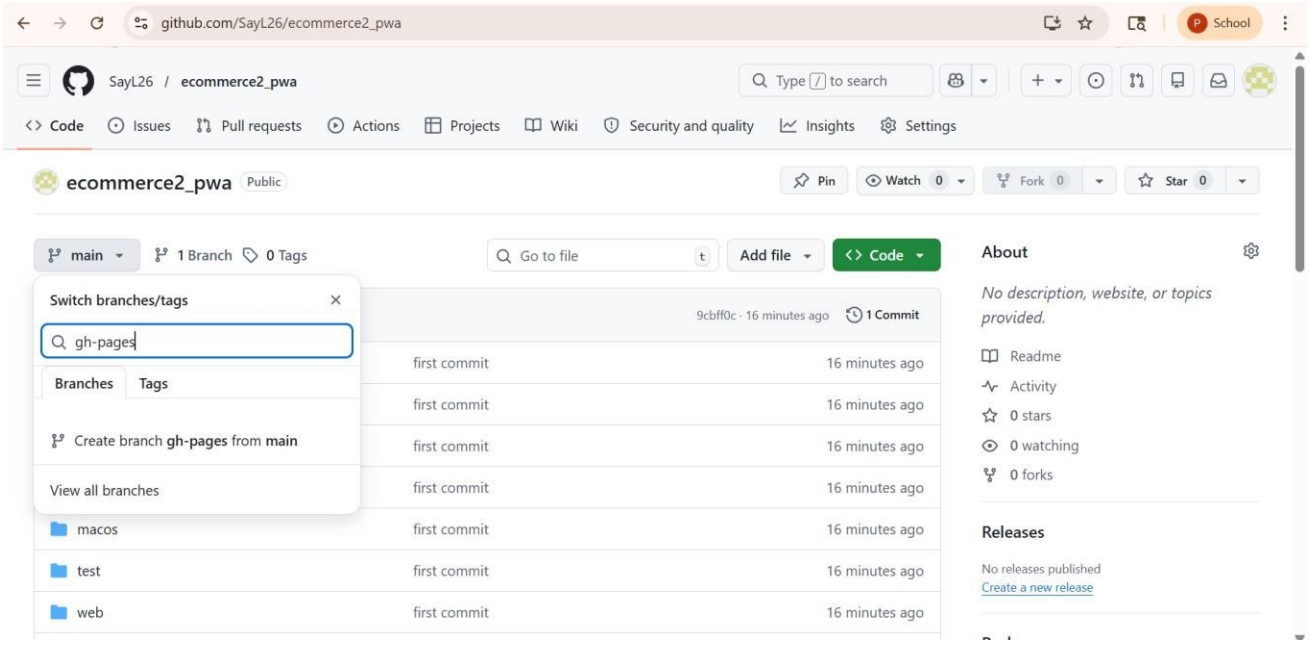
git push -u origin main

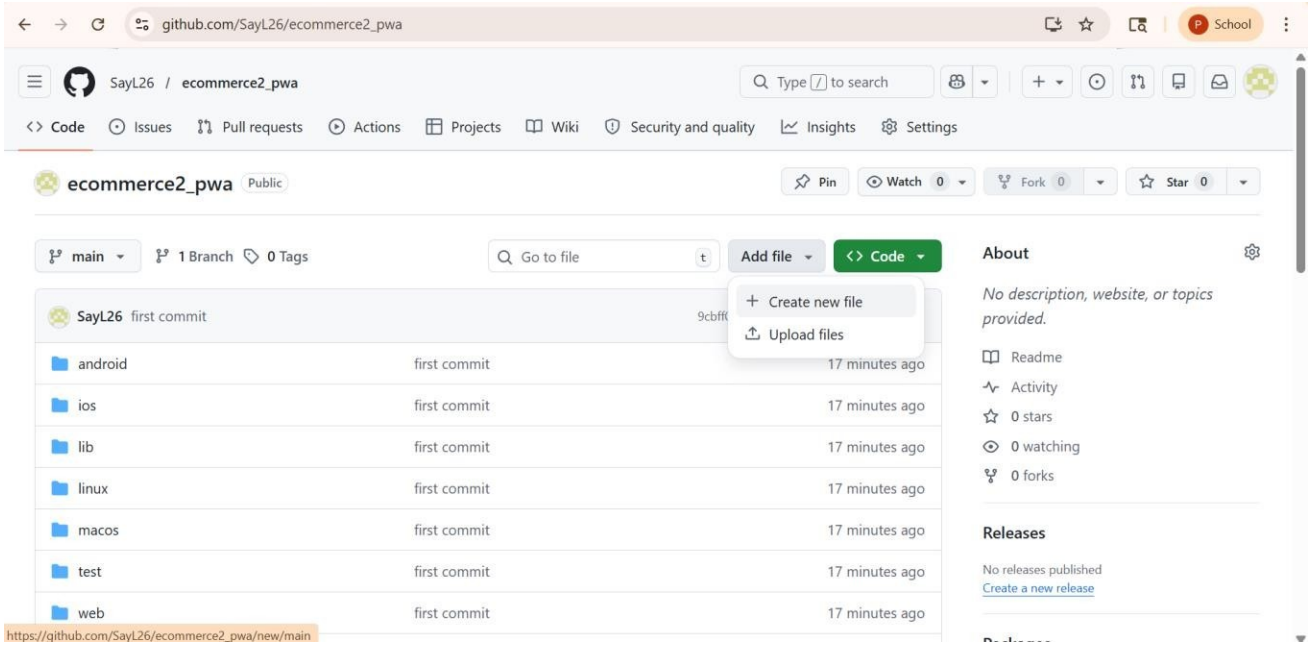
```
PS C:\Users\Lenovo\OneDrive\Desktop\MAD_25_26\Exp-8\flutter_firebase\new_ap
plication\ecommerce_pwa\ecom> git remote add origin https://github.com/SayL
26/ecom.git
PS C:\Users\Lenovo\OneDrive\Desktop\MAD_25_26\Exp-8\flutter_firebase\new_ap
plication\ecommerce_pwa\ecom> git branch -M main
>> git push -u origin main
Enumerating objects: 179, done.
Counting objects: 100% (179/179), done.
Delta compression using up to 8 threads
Compressing objects: 100% (148/148), done.
Writing objects: 100% (179/179), 266.07 KiB | 5.78 MiB/s, done.
Total 179 (delta 19), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (19/19), done.
To https://github.com/SayL26/ecom.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

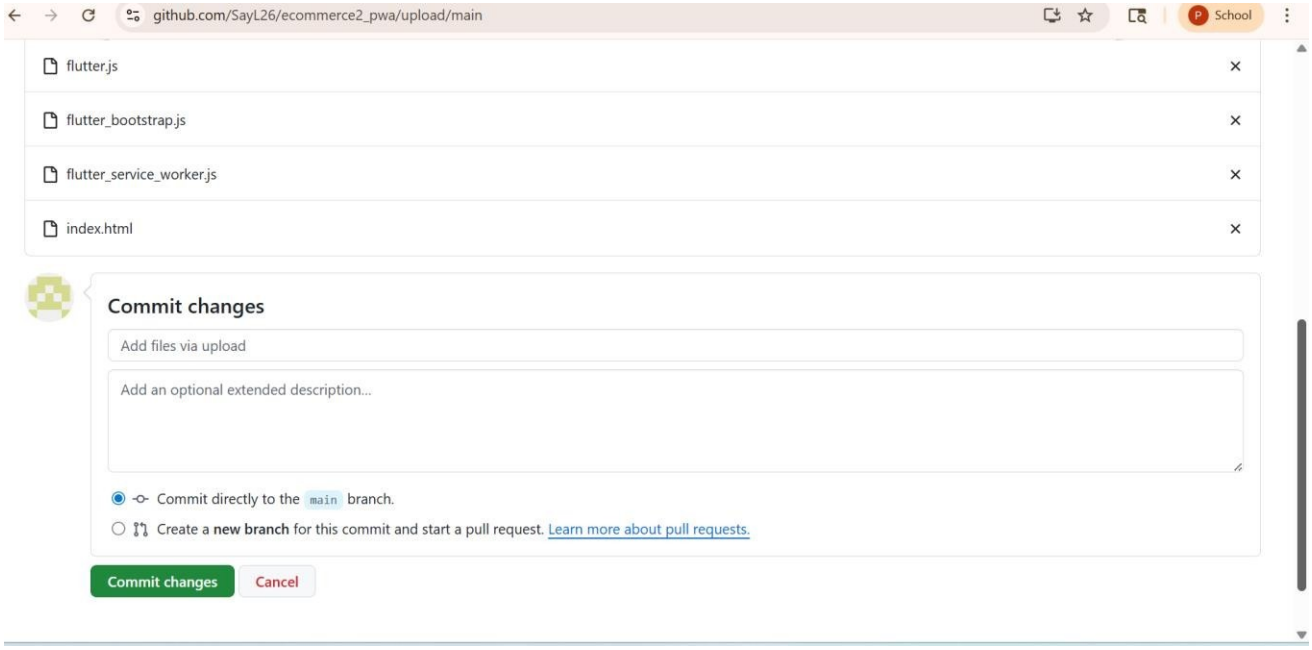
Step 9: Upload Files



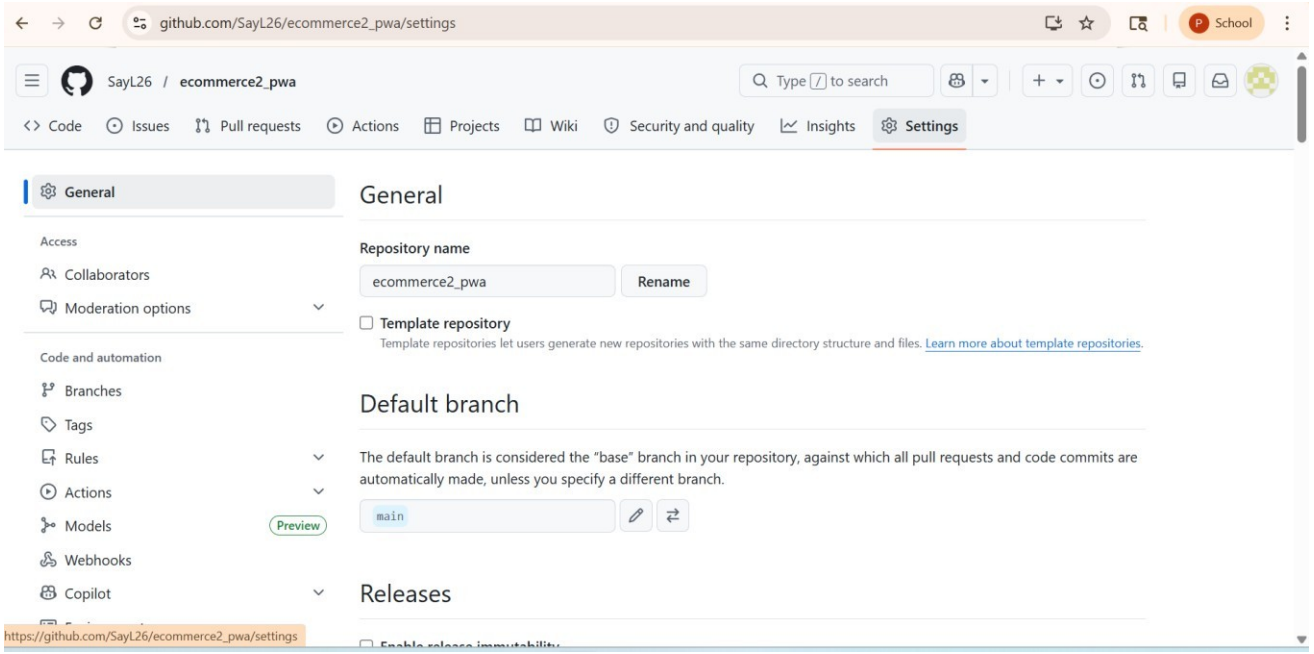
1. **Inside your new repository**
2. **Click "Add file" → "Upload files"**
3. **Open your project folder from your computer**
4. **Drag and drop ALL files into GitHub**
5. **Scroll down**
6. **Click "Commit changes"**

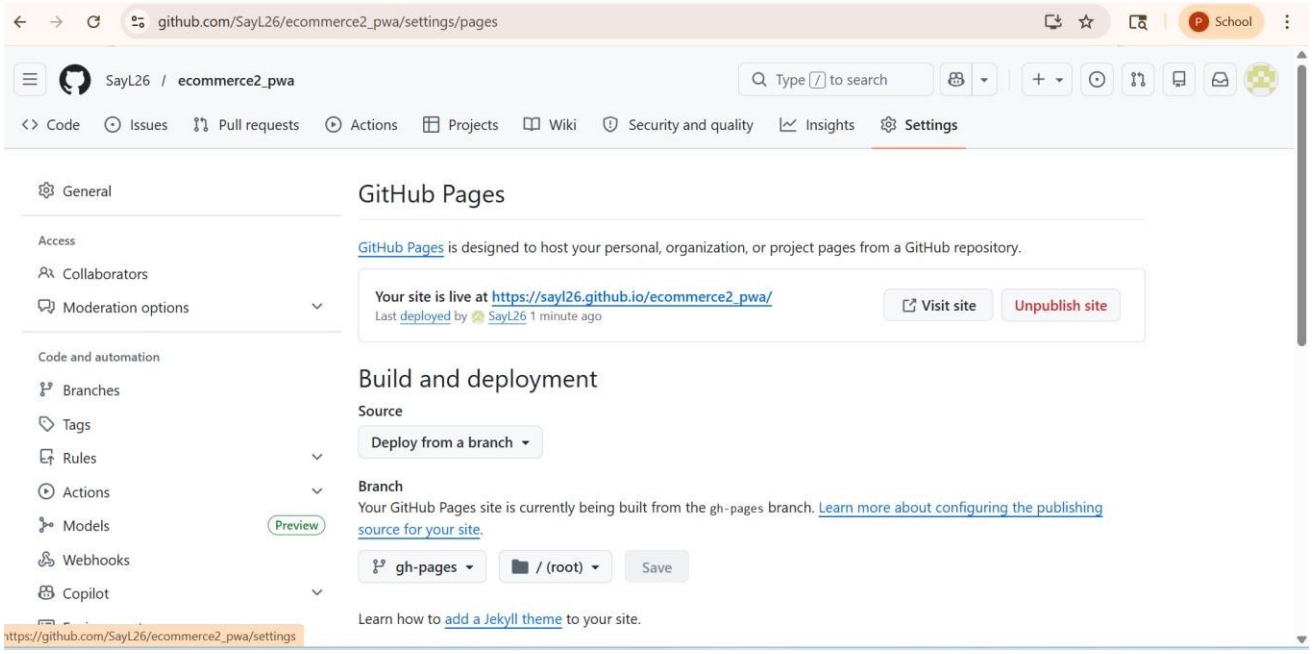






Step 12: Enable GitHub Pages


github.com/SayL26/ecommerce2_pwa/settings
SayL26 / ecommerce2_pwa
Type / to search
Code
Issues
Pull requests
Actions
Projects
Wiki
Security and quality
Insights
Settings
General
Access
Collaborators
Moderation options
Code and automation
Branches
Tags
Rules
Actions
Models
Preview
Webhooks
Copilot
General
Repository name
ecommerce2_pwa
Rename
Template repository
Template repositories let users generate new repositories with the same directory structure and files. Learn more about template repositories.
Default branch
The default branch is considered the "base" branch in your repository, against which all pull requests and code commits are automatically made, unless you specify a different branch.
main
Releases
https://github.com/SayL26/ecommerce2_pwa/settings


github.com/SayL26/ecommerce2_pwa/settings/pages
SayL26 / ecommerce2_pwa
Type / to search
Code
Issues
Pull requests
Actions
Projects
Wiki
Security and quality
Insights
Settings
General
Access
Collaborators
Moderation options
Code and automation
Branches
Tags
Rules
Actions
Models
Preview
Webhooks
Copilot
GitHub Pages
GitHub Pages is designed to host your personal, organization, or project pages from a GitHub repository.
Your site is live at https://sayl26.github.io/ecommerce2_pwa/
Last deployed by SayL26 1 minute ago
Visit site
Unpublish site
Build and deployment
Source
Deploy from a branch
Branch
Your GitHub Pages site is currently being built from the gh-pages branch. Learn more about configuring the publishing source for your site.
gh-pages
/ (root)
Save
Learn how to add a Jekyll theme to your site.
https://github.com/SayL26/ecommerce2_pwa/settings

Conclusion:
Deploying an eCommerce Progressive Web App on GitHub Pages ensures easy, cost-effective hosting with global accessibility. It enables users to experience fast, reliable, and app-like shopping directly through their browsers.